

IBM – Capstone: Today and Future of Coding

Ali Can TURAN

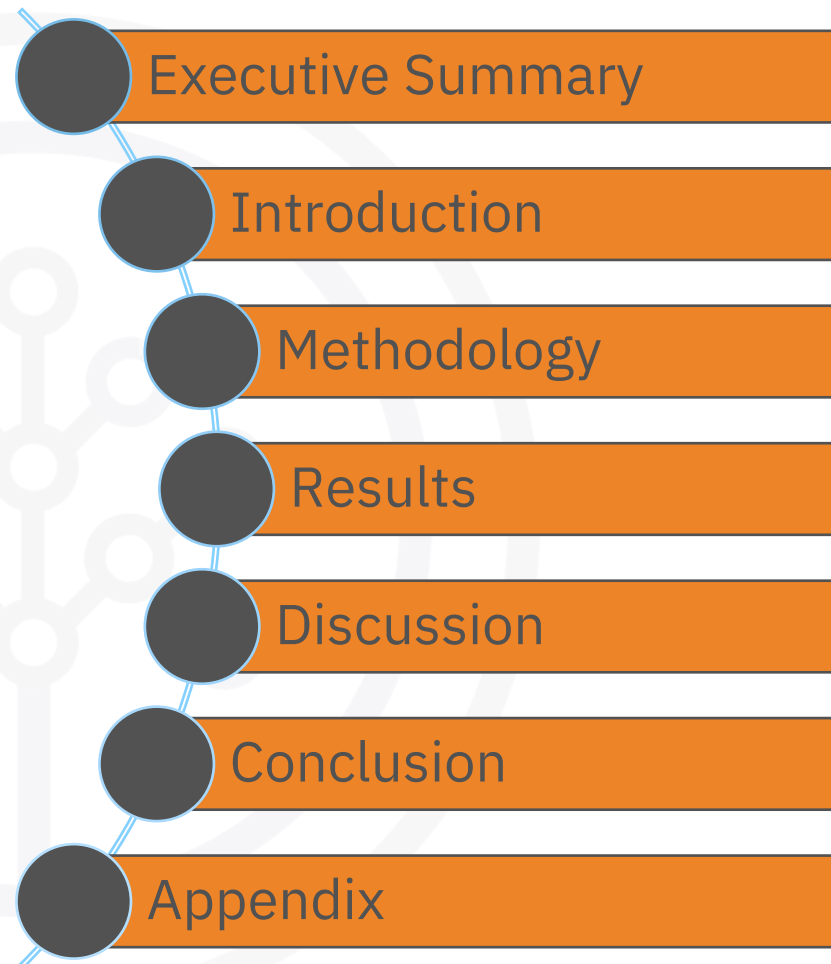
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OUTLINE





EXECUTIVE SUMMARY

- **Washington DC:** the largest quantity of available jobs on the market.
- **Full-time employees:** mainly bachelor's degree holders.
- **75% reported earning 100,000 \$ or less annually.**
 - **500,000 \$ or more:** more than half are 25-34 years old.
 - **The most stretched and the highest:** India and USA respectively. ([Appendix for patterns in the top 5 countries.](#))
 - **The highest** for engineering managers and **the lowest** for academic researchers EXCLUDING students.
- **Preferred tools and systems:**
 - Windows: (35%)
 - Amazon Web Services: (32%)
 - JavaScript: (60%)
 - PostgreSQL: (34%)
 - VSC: (72%)
 - React: (11%)
- **Job Satisfaction:**
 - Decreases gradually after 3 years of experience and increases considerably after 20 years of work experience.



INTRODUCTION



Aim

- To extract popular coding environments and tools,
- To explore their relations to education level, compensation and satisfaction levels at job.

Audience

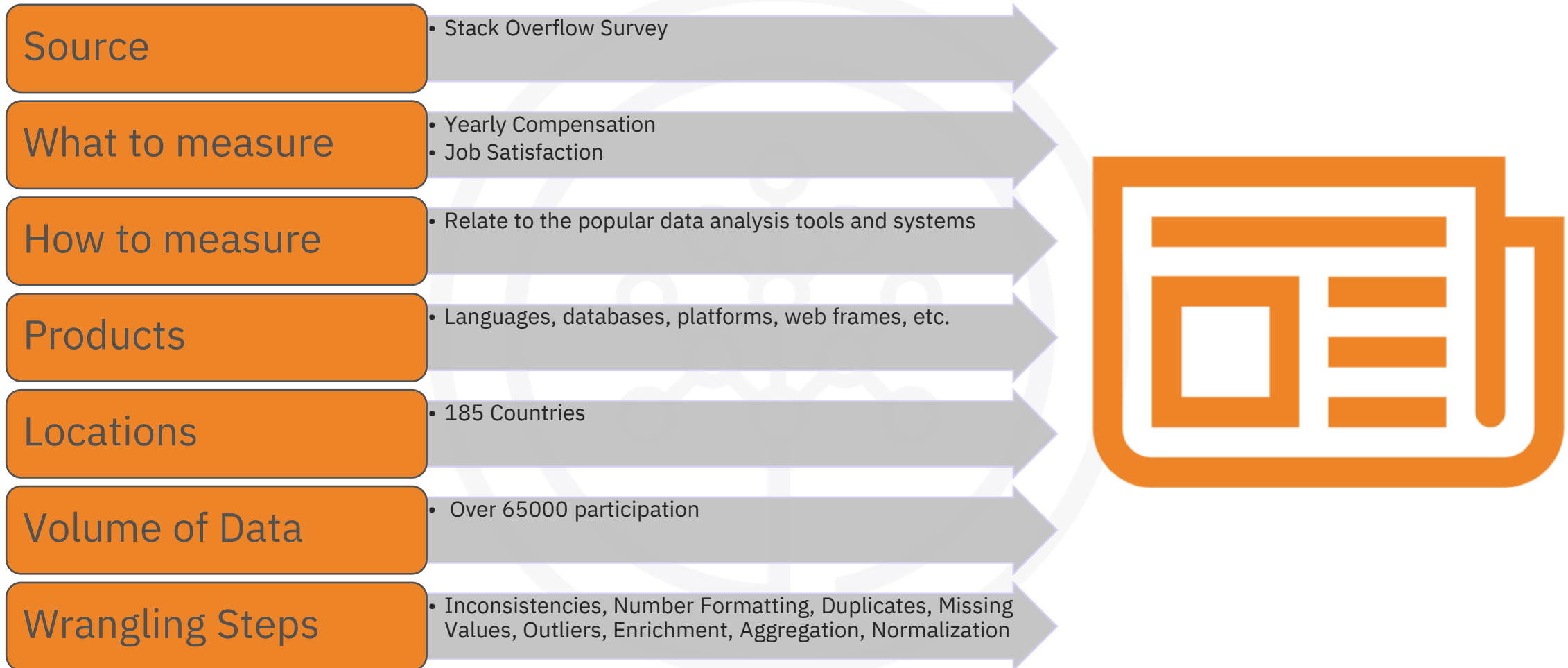
- Anyone with the interest in data analysis and coding.

Value

- Earn insights for the future's data analysis sphere to have optimum compensation and satisfaction at job.



METHODOLOGY



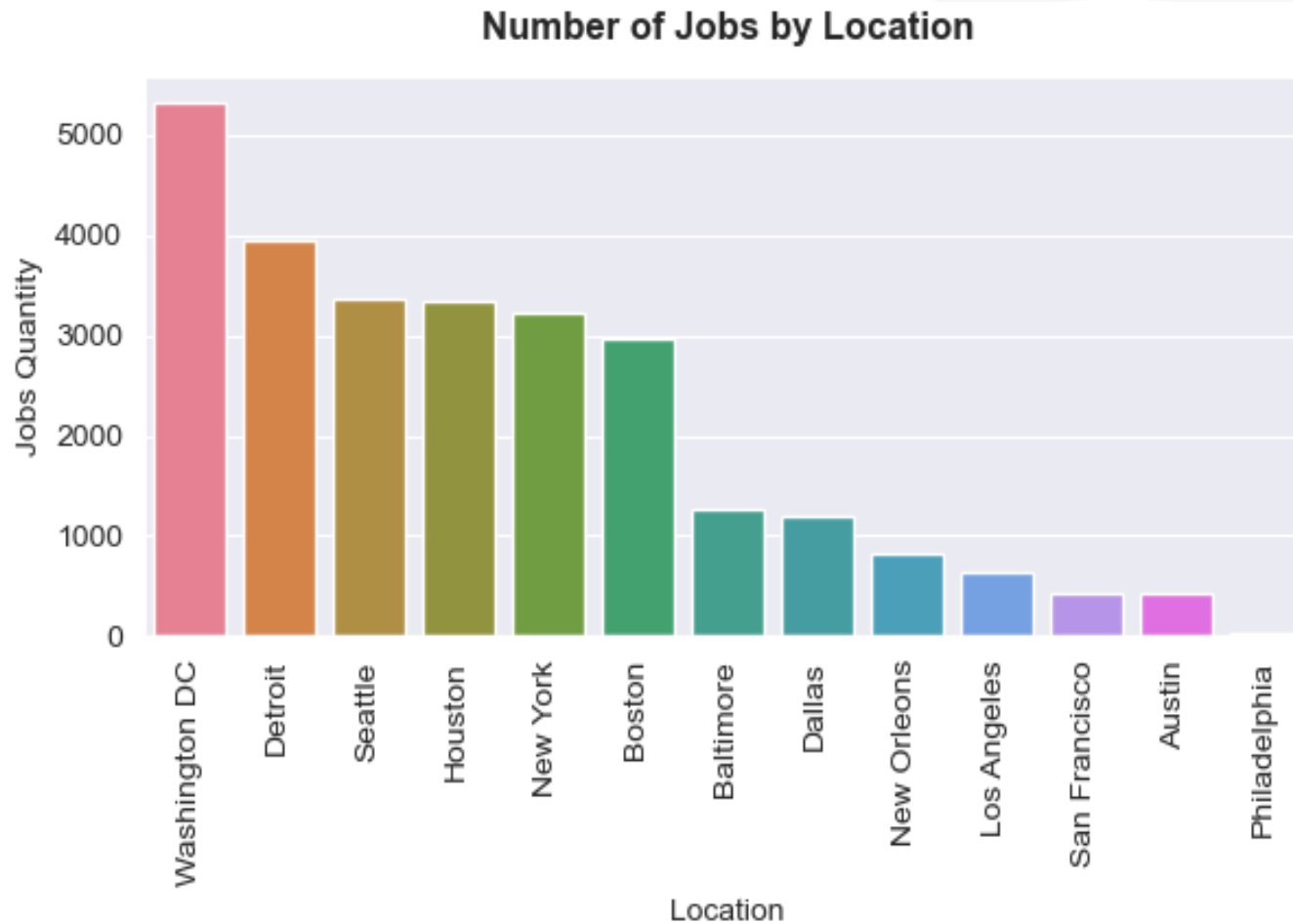
Results



<https://github.com/ali-can-turan/ibm-capstone>



JOB POSTINGS



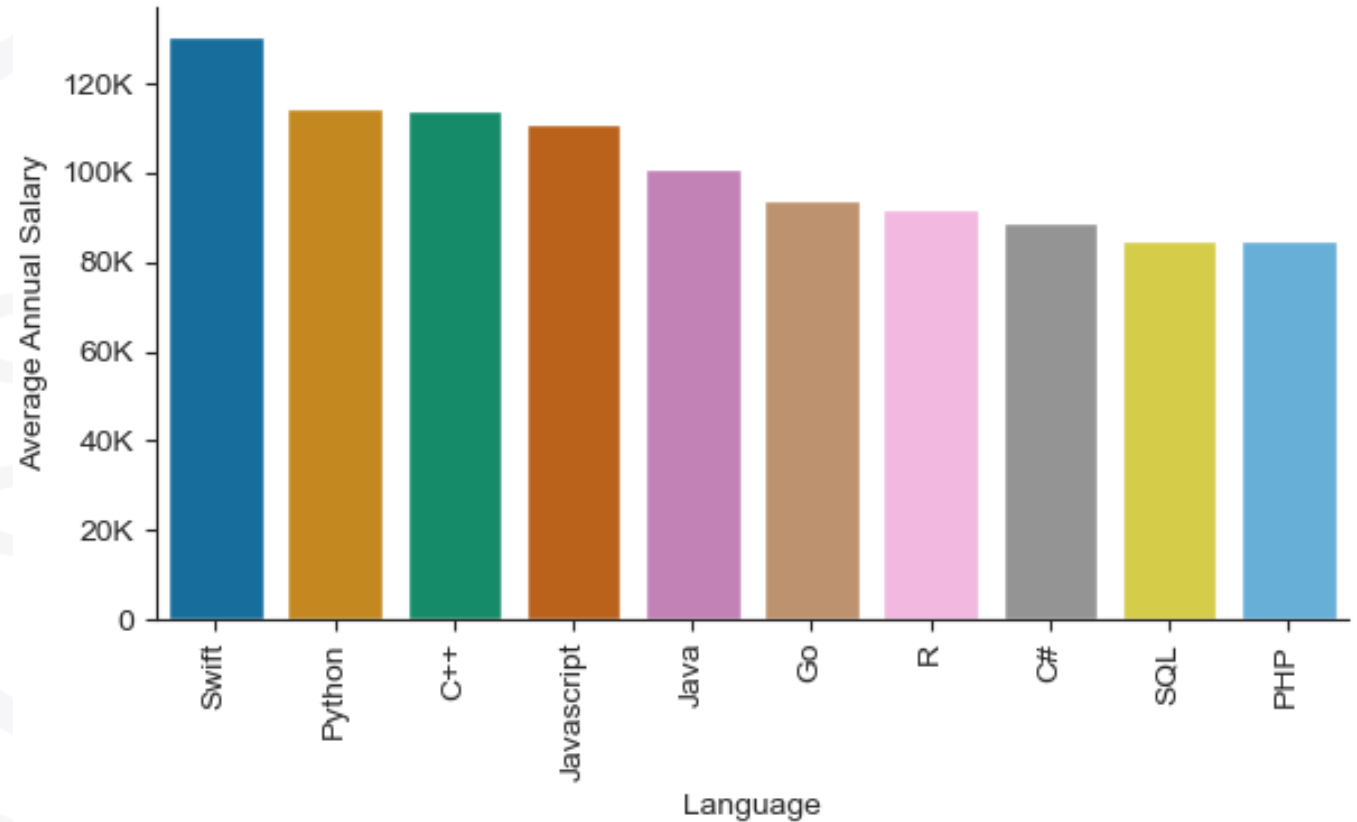
As seen from the chart, «Washington DC» is the province that has **the largest number of jobs** available, getting ahead of its nearest rival «Detroit» by a margin of **more than 35%.**

POPULAR LANGUAGES

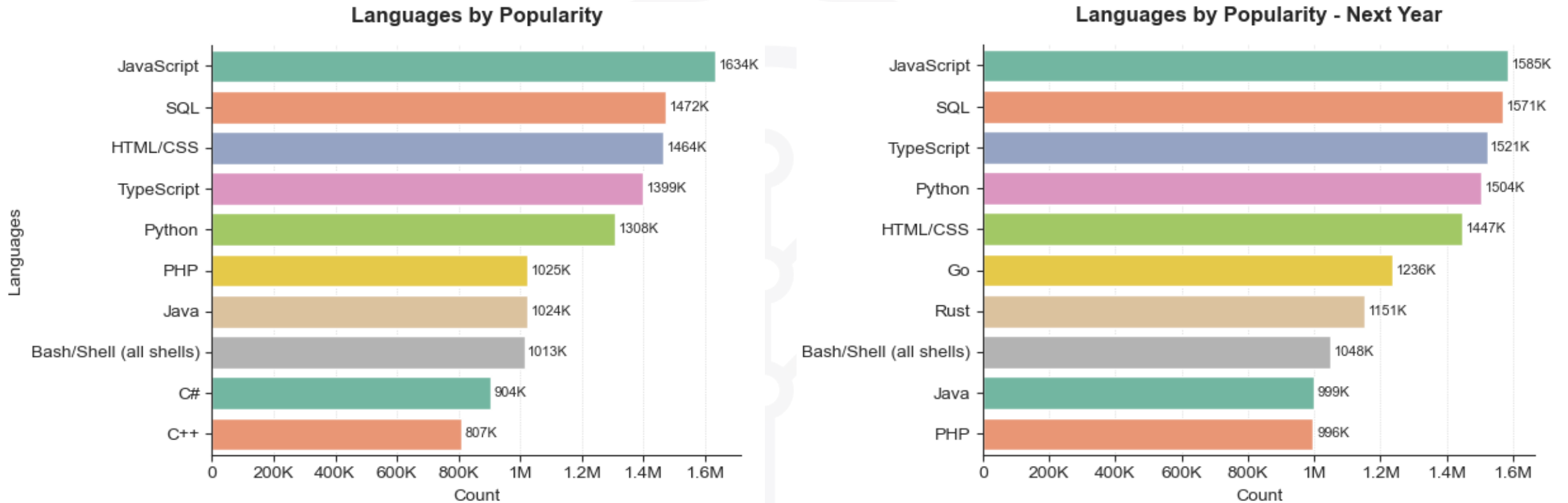
As seen from the chart, «Swift» is the programming language of which the users earn the highest salary: **130K**, «Python» being the second with **115K**.



Average Annual Salary by Language



PROGRAMMING LANGUAGE TRENDS



- Top 5 in either graph are comprised of the same languages. However, **HTML/CSS** seems to be going to lose ground next year.
- Next year, it seems that **C# and C++** will drop from the top 10 and that **Go and Rust** will get in the list from a high ground overtaking **PHP, Java and Shell languages**.

PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

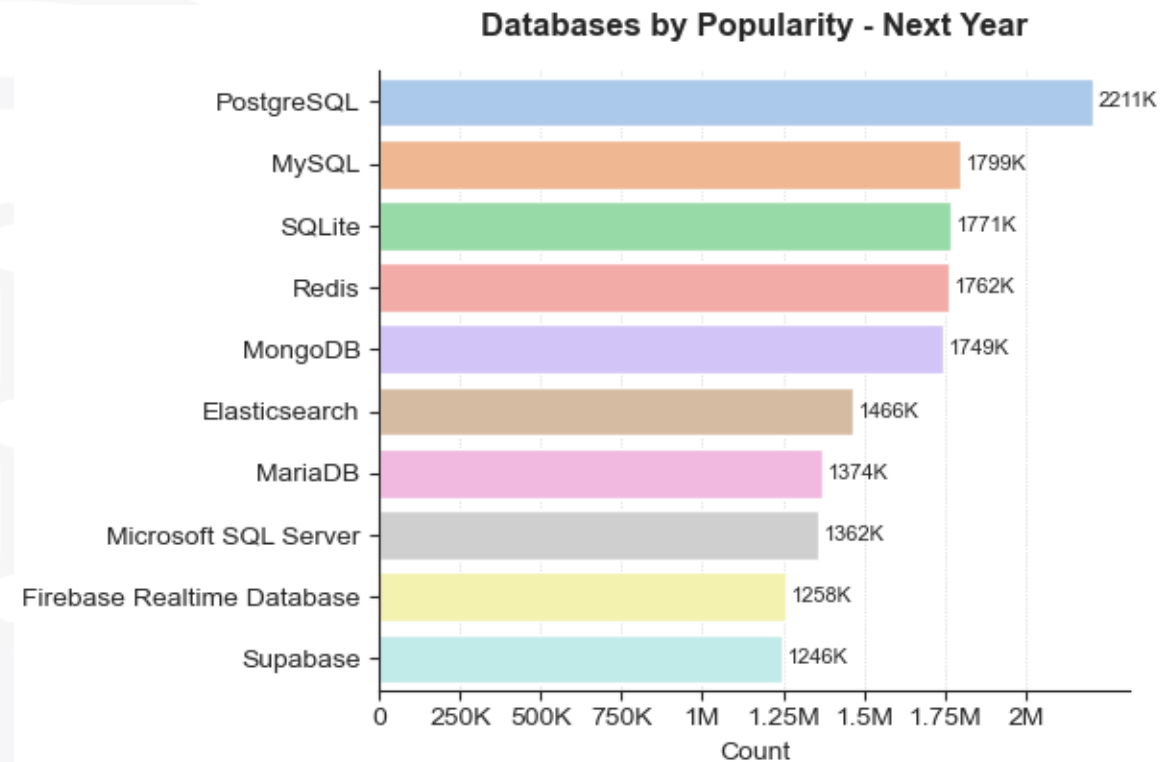
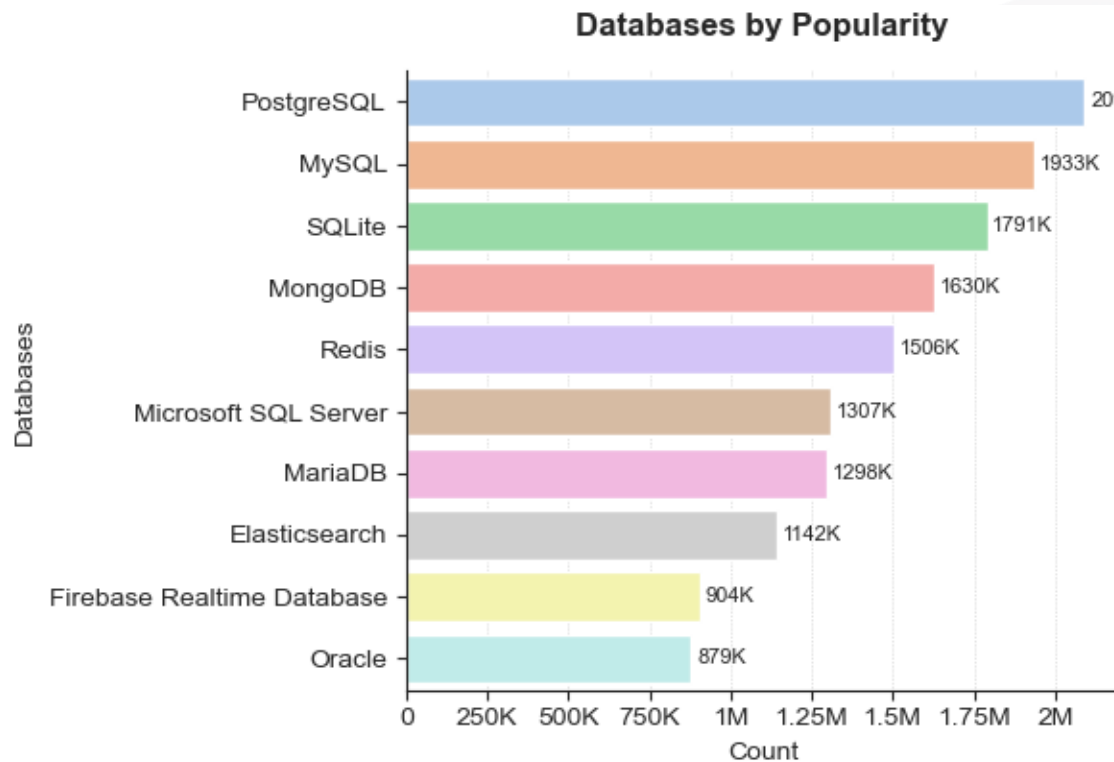
- JavaScript is the market leader.
- SQL is the second.
- TypeScript is gaining ground.
- Python is at the first 5.
- Go and Rust are fast-moving.

Implications

- Web page/app developers are dominant, at least, in the survey.
- No rival in the realm of database manipulation.
- More significance on type, compared to loosely-typed JS.
- Seems not to get outdated in the near future.
- Will see new breakthroughs from these languages



DATABASE TRENDS



- There seems to be no difference in database popularity in the near future for the top 3: **PostgreSQL, MySQL, SQLite**.
- On the other hand, slight attack in **Redis and Maria DB** and moderate gain in **Elasticsearch** is visible.

DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL is the market leader.
- Decrease in MySQL is overt.
- MS SQL Server not catching up.

Implications

- Seems not to get outdated in the near future.
- The rivalry between from the second to the fourth is more evident.
- Rivalry at the top, new trends at the bottom; MS SQL Server not in either category

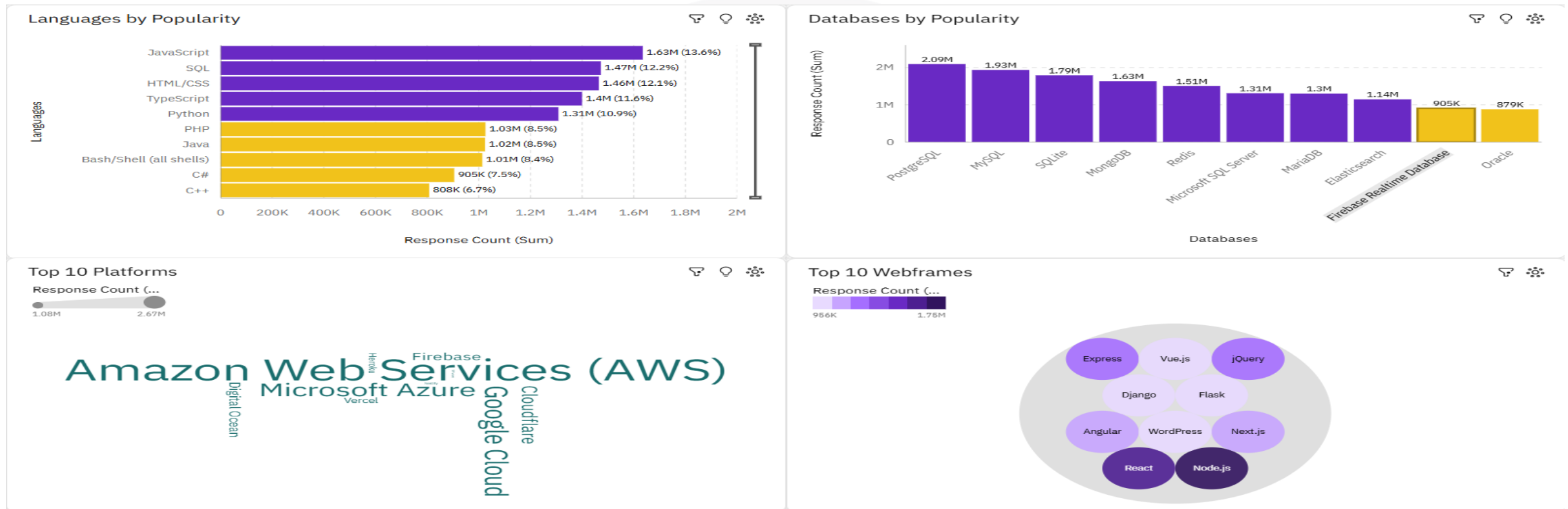
DASHBOARD



<https://github.com/ali-can-turan/ibm-capstone>



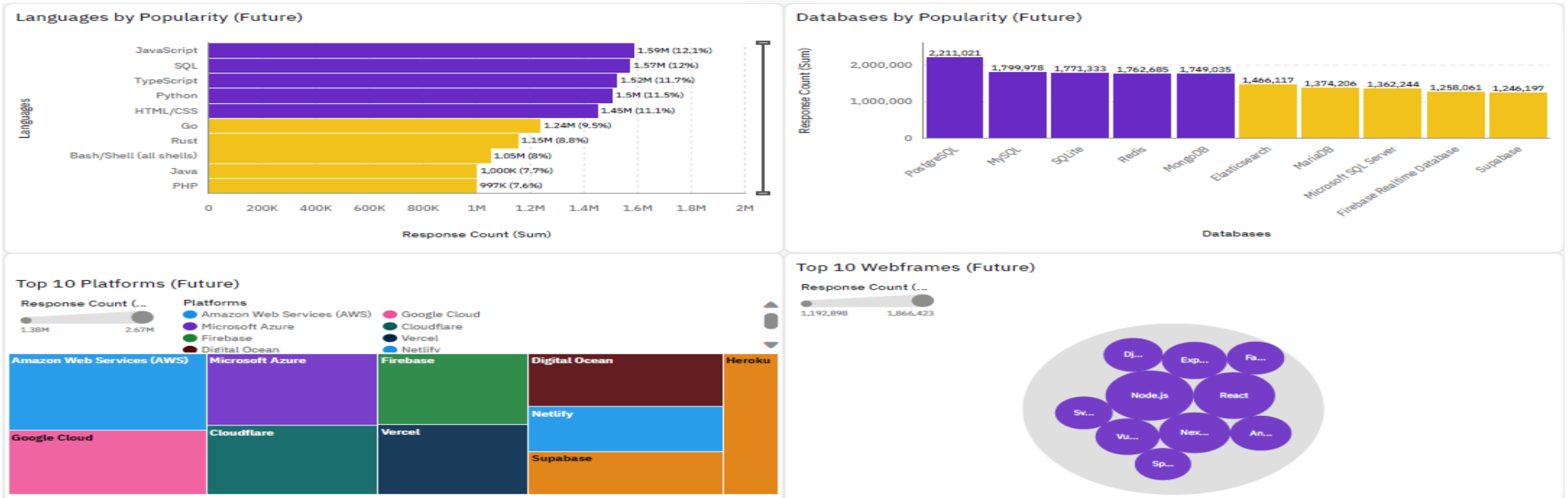
DASHBOARD TAB 1: Current Technology Usage



- **JS** stands out as the leader by a margin of **10%** from the closest rival: **SQL**.
- Rivalry between the databases is more eminent compared to the languages, which means no single brand should be overlooked.
- **Amazon Web Services** is visibly the most popular platform leaving the others way behind.
- **Node.js** seems to continue its prominence in the back-end with speed advantages.



DASHBOARD TAB 2: Future Technology Trends



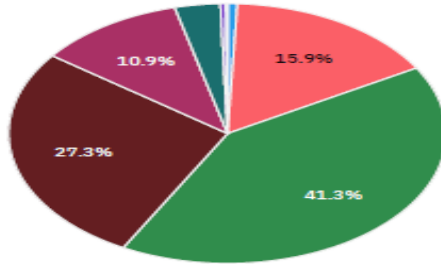
- **JS** is no longer the by-far-leader. Rivalry is now eminent between the **first 5**.
- There seems that **PostgreSQL** will solidfy its ground as the first. But rivalry is getting visible between the **4 chasers**.
- **Amazon Web Services** seems to protect its place. But now rivals are following more closely.
- There seems to be no change in the popularity hierarchy within web frames.



DASHBOARD TAB 3: Demographics

Respondent Distribution by Age

Under 18 years old 18-24 years old 25-34 years old 35-44 years old 45-54 years old
55-64 years old 65 years or older Prefer not to say



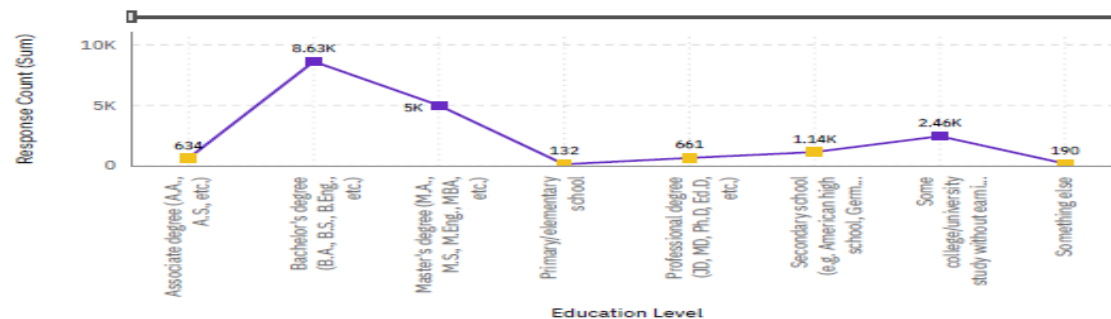
Respondent Distribution by Country

Count (Sum)



© Mapbox © OpenStreetMap

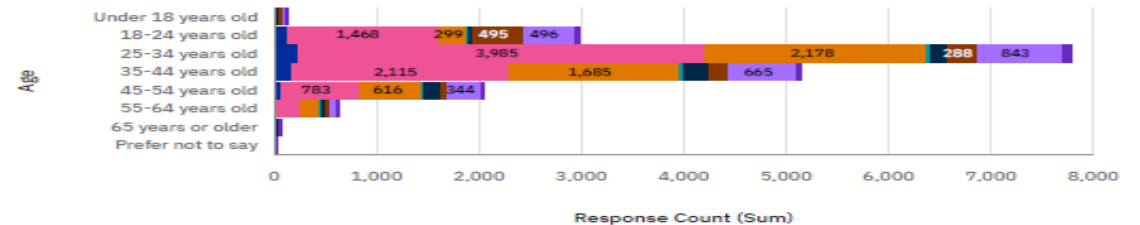
Respondent Distribution by Education Level



Respondent Distribution by Age, Classified by Education Level

EdLevel

Associate degree (A.A., A.S., etc.) Bachelor's degree (B.A., B.S., B.E., etc.) Master's degree (M.A., M.S., M.Eng., etc.)
Primary/elementary school Professional degree (JD, MD, Ph.D., etc.) Secondary school (e.g., American high school, etc.)
Some college/university study without earning a degree Something else

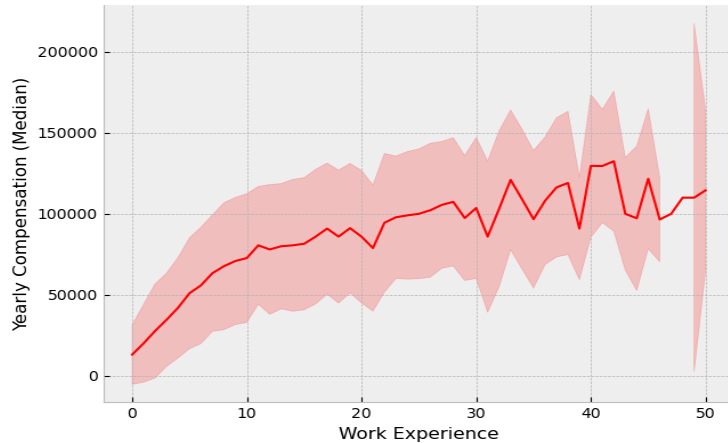


- 35-44 year-olds are only beaten by the 18-24 and 35-44 year-olds combined.
- Majority of respondents are from **the USA**.
- More than **45%** of the respondents are **bachelor's degree** holders, **EXCLUDING** no-answers. (Appendix for «Employment vs Education Level» graph)
- 25-34 year-olds have more bachelor's degree holders than any other age group both by **quantity and rate**.



Yearly Compensation – Job Satisfaction

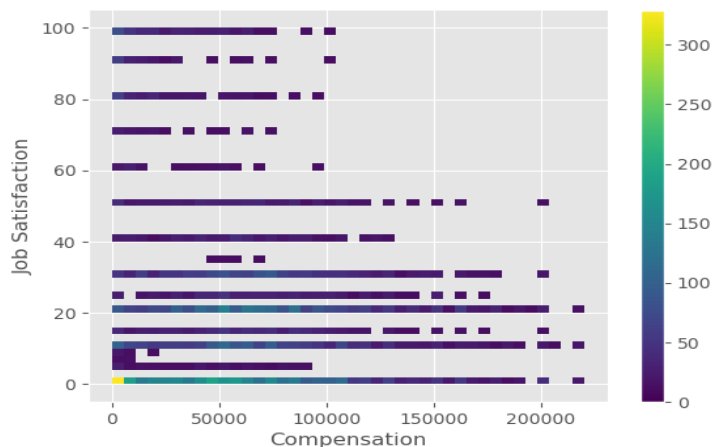
Yearly Compensation (Median) by Work Experience



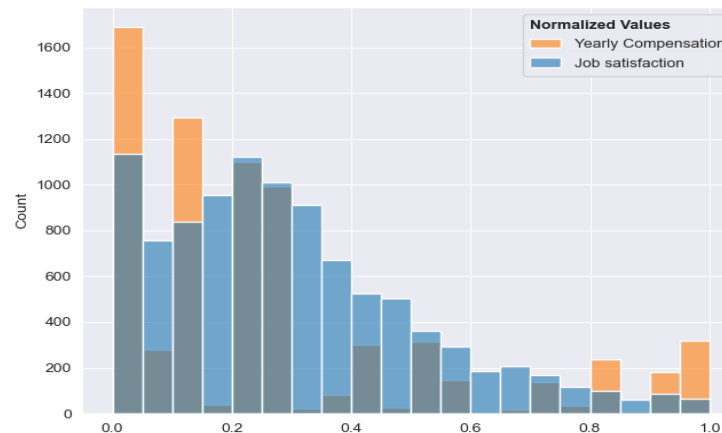
Job Satisfaction (Avg.) by Work Experience



Compensation vs. Job Satisfaction



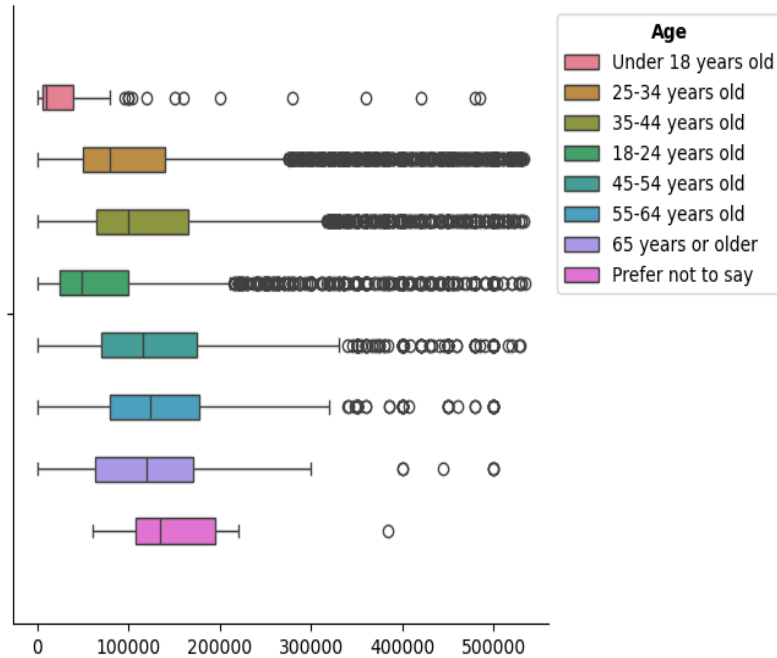
Yearly Compensation and Job Satisfaction Distribution (Normalized)



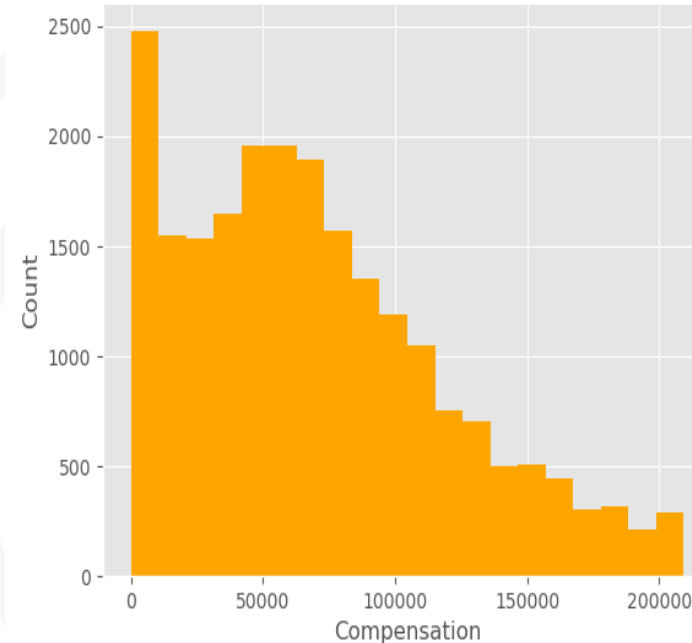
- As with the job satisfaction, annual compensation also fluctuates considerably for respondents with more than **32 years of work experience**.
- More than half of the coders participating in the survey did not provide an answer for the questions about **“Job Satisfaction”** and **“Yearly Compensation”** as well as AI related ones.
- The correlation between the **annual compensation** and **job satisfaction** is positive but moderate.

Yearly Compensation

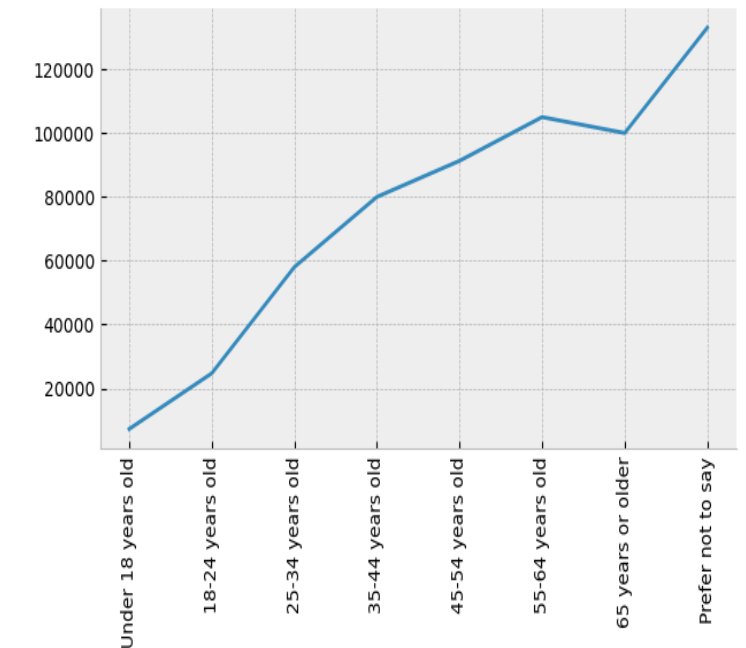
Yearly Compensation Distribution by Age Group



Yearly Compensation Distribution



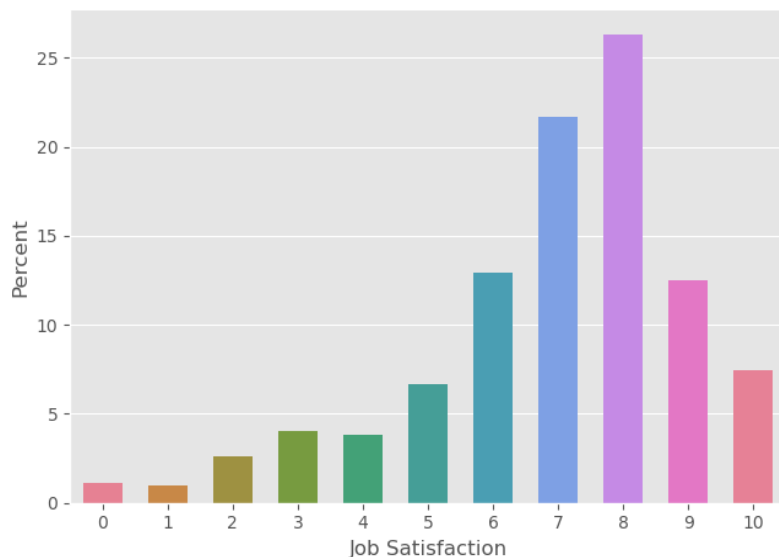
Compensation Change Between Age Categories



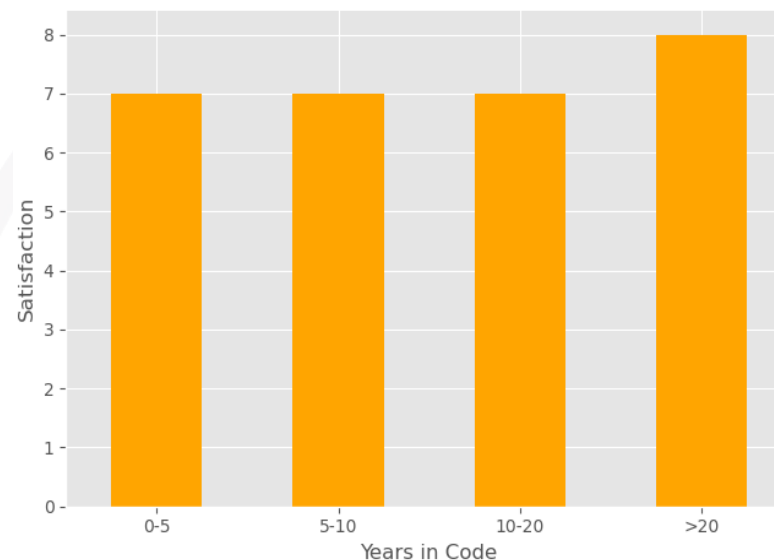
- The highest and narrowest yearly compensation comes from the respondents who do not provide their age, EXCLUDING outliers.
- Yearly Compensation data is **right-skewed**, but there is a peak of nearly **2500 respondents** with **10000 dollars or less** annual income. Nearly **80%** of those are between **18-34 years old**.
- Total compensation seems to be almost **linearly and positively correlated** with age and work experience, except that the increase halts and reverts to decrease **after the age of 65**.

Job Satisfaction

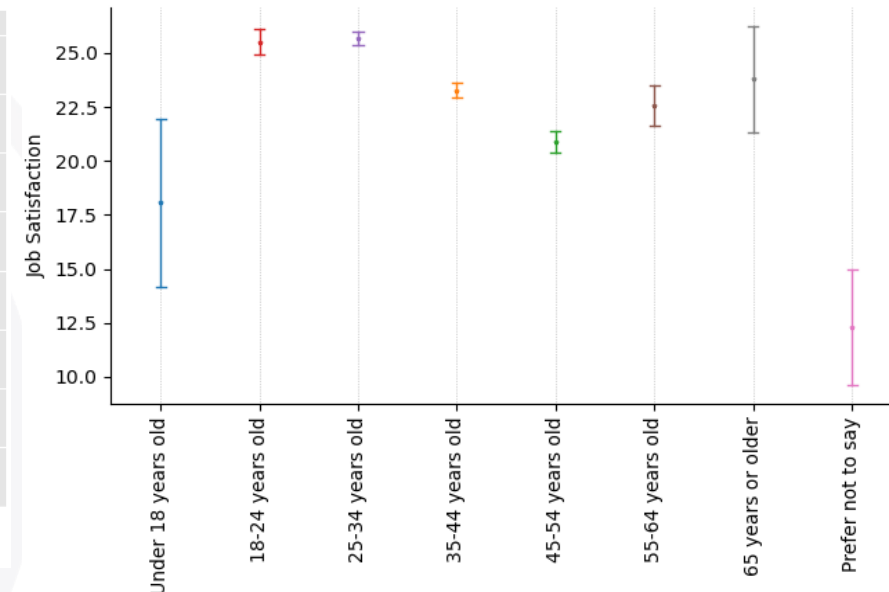
Job Satisfaction Distribution



Job satisfaction by Years in Code



Age vs. Job Satisfaction



- Job satisfaction mean is **8 over 10**. This is mainly the result of **the unemployed** and the ones with the **20 years or more work experience**.
- Job satisfaction average is at its highest (slightly more than 25 over 100) for the respondents of **18-34 years old**.
- Job satisfaction is at its low ebb (around 15%) for respondents who indicated Julia and at its peak (around 28%) for respondents who indicated Solidity as languages to work with.
- Respondents with the most job satisfaction levels use Raven DB database and/or Gatsby web frame, while some others with the least job satisfaction levels use Couch DB database and/or Django web frame.



DISCUSSION

Jobs quantity in Washington DC is **65%** more than that in New York...

- ...THOUGH the latter is **two and a half fold** more populated than the former.

C++ is one of four languages whose users earn **more than 100K** annually...

- ...DESPITE its unpopularity, **not being in top 10**.

Front-end developers earn **20% less** compared to the full-stack and back-end developers...

- ...WHILE they constitute **one-nineth** of the respondents of mentioned 3 categories.

High earners are usually **24-35 year-olds with a bachelor's degree**...

- ...THOUGH the highest job satisfaction is from the unemployed

Swift is the language whose users earn the most on average...

- ...BUT the reason for this is not definite.

Rivalry between database brands are overt...

- ...THOUGH PostgreSQL seems not to bother as a leader.



OVERALL FINDINGS & IMPLICATIONS

Findings

- HTML/CSS is losing ground...
- SQL is the market leader...
- Rivalry between frameworks are extremely vivid...
- PostgreSQL is way ahead of rivals...
- Amazon Web Services will keep the ground...
- Highest and the lowest job satisfaction average from 2 categories...

Implications

- DUE TO frameworks that ease the work such as streamlit or dash.
- DUE TO unavailability of the rivals out there.
- SO MUCH SO THAT React is the leader only with 11% of the market share.
- AND this seems not to change in the near future.
- BUT the threat from the MS and Google services will grow bigger.
- INTERESTINGLY the unemployed and the retired respectively.



CONCLUSION



100K will continue to be the ballpark average for the yearly compensation, with 250K being the virtual top EXCLUDING outliers.



Job satisfaction will still base on work experience, work modes and work stress rather than compensation.



JavaScript, Python, SQL, TypeScript, PostgreSQL, Amazon Web Services, Node.js, React will be the leaders of the near future, not without threatening rivals. Go and Rust are the new age within programming languages



MySQL should be watched carefully in the next years as the popularity decrease might turn into chronic.



HTML/CSS will continue to lose ground as simpler frameworks allow to build esthetic and interactive web pages / apps.

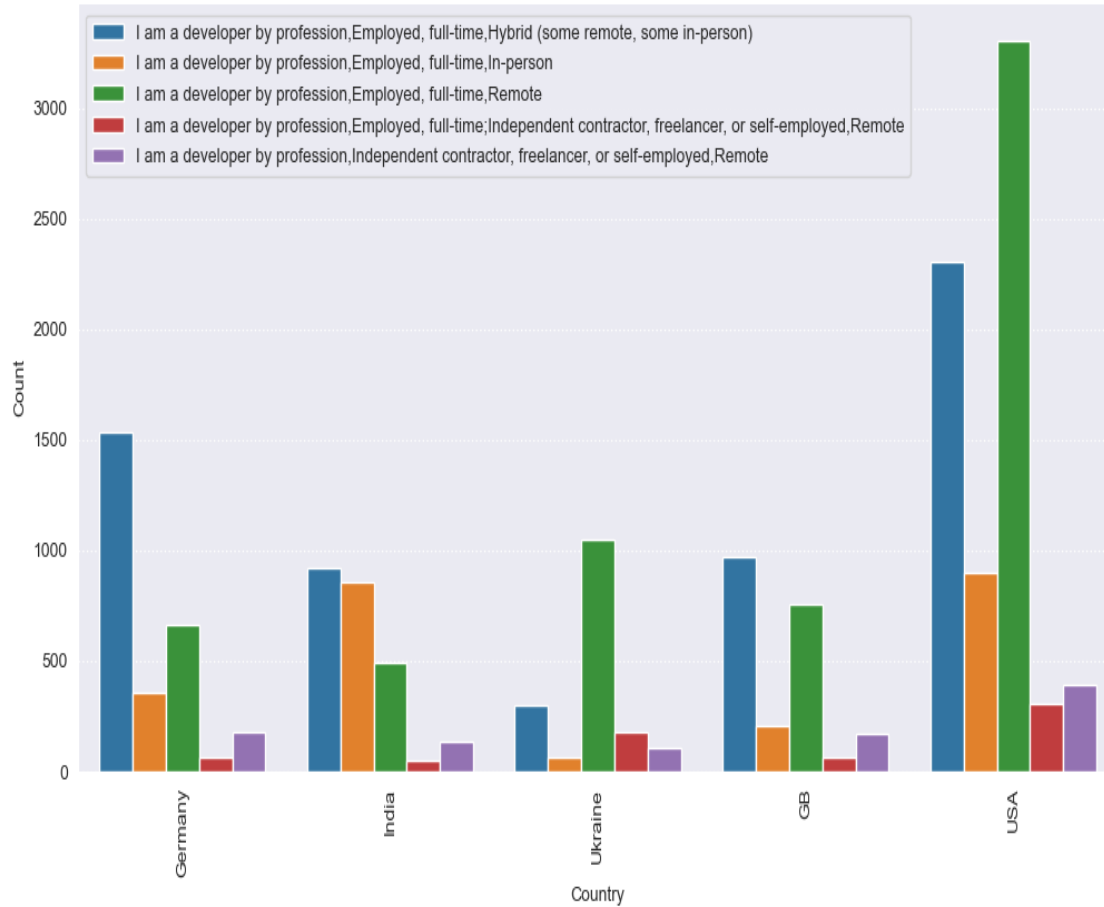


University degree will still continue to bring up more job opportunities, but often with in-person work mode.

APPENDIX



Top 5 Countries with top 5 Patterns



Employment Type

Employment vs Education Heatmap

